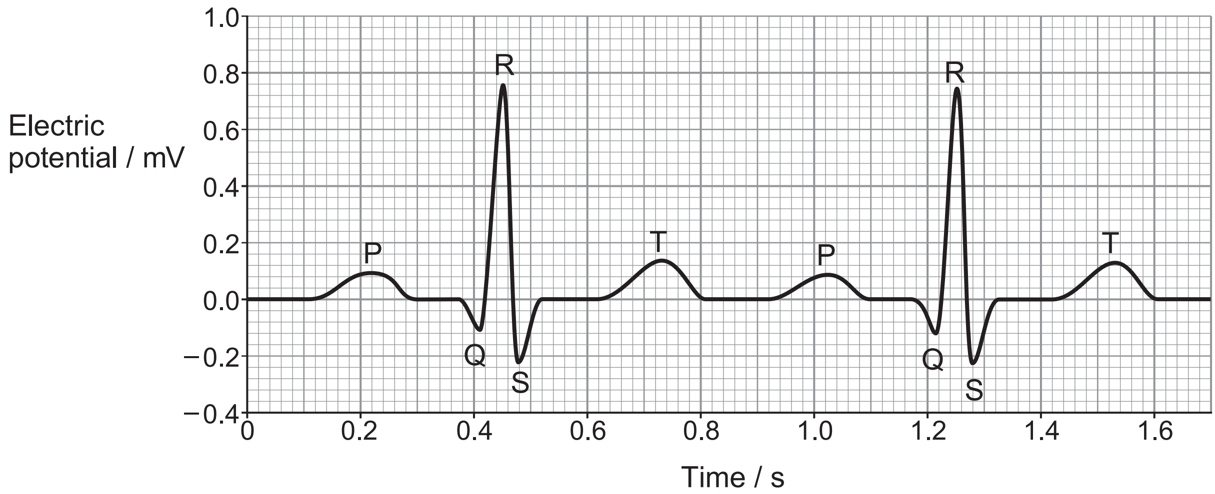
Examiner only

5. An ECG is a test that can be used to check the heart's rhythm and electrical activity. Sensors attached to the skin are used to detect the electrical signals produced by the heart each time it beats. The graph below shows part of a trace from a healthy person at rest.



(a) (i) What does the abbreviation ECG represent? [1]

....

(ii) Calculate the heart rate of the person in beats per minute (bpm). [1]

Heart rate = ... bpm

(iii) Explain the events occurring during;

I. The P wave. [3]

....

....

....

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II. The QRS complex. [3]

....

....

....

....

III. The T wave. [2]

....

....

....

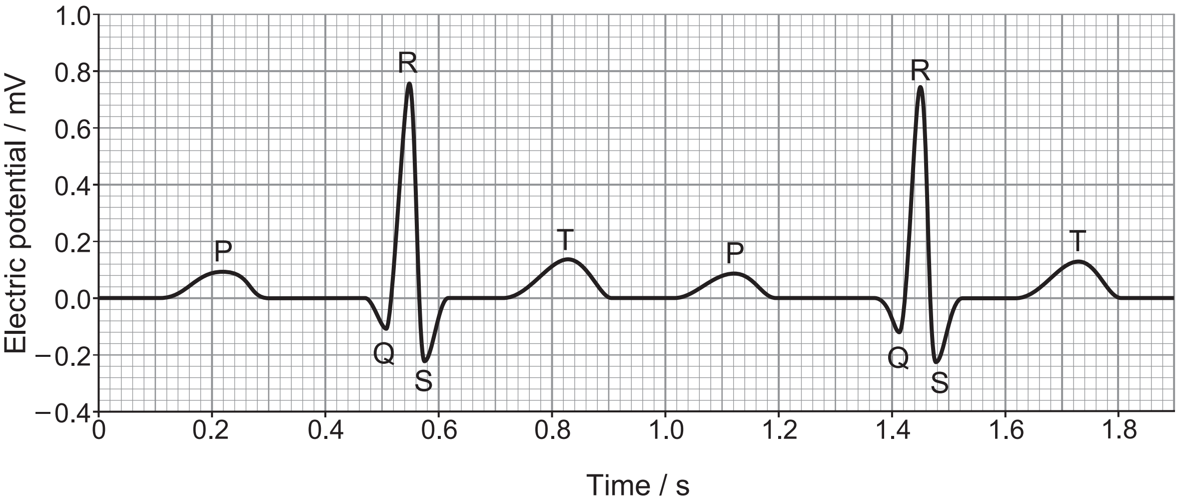
(b) During exercise there is little change to the lengths of the P wave, QRS complex, or T wave. Describe and explain how the distance between consecutive P waves would differ in a person taking exercise. [2]

....

....

....

(c) The ECG trace below illustrates an abnormality known as a First Degree Heart Block.



(i) On the graph above **circle one** region of the ECG where this abnormality occurs. [1]

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(ii) Conclude which region of the conducting tissue of the heart is affected by a First Degree Heart Block. [1]

....

(iii) Suggest the effect that a First Degree Heart Block would have on the functioning of the heart. [1]

....

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